

Steel Beam Estimator

Version 2 - Pipeline Workflow Document

Phases A through C.5.2 | June 2026 | Pre-Phase D

1. Executive Summary

Version 2 extracts structural beam data from two AutoCAD DXF sources: the framing plan (beam marks and centreline geometry) and the reinforcement details sheet (section headers, grid layout, and detail sketches). The pipeline runs from framing extraction (Phase A) through reinforcement headers (Phase B), spatial grid cells (Phase C), sketch detection, and sketch-to-header ownership validation (Phases C.5 through C.5.2).

Version 1 is frozen. All development occurs only inside Version2/.

Current sample drawing results

Metric	Value
Unique beams (framing)	18 (B1-B18)
Header occurrences	24
Sketches detected	38
Ownership assignments	38
Ownership status	PASS
Audit status	PASS_WITH_WARNINGS

2. Input Data

File	Path	Purpose
Framing plan	data/framing/Beam_FramingPlan.dxf	Labels + STR-BEAM geometry
Reinforcement	data/reinforcement/Beam_ReinforcementDetails.dxf	SEC TEXT + details

3. Pipeline Phases

Phase A - Framing Plan Extraction

Match B{n}(WxD) labels to STR-BEAM centreline segments. Flatten INSERT blocks. Outputs: framing_beams.json (18 beams), framing_validation.json.

Modules: src/framing/framing_beam_extractor.py, src/parser/dxf_flattener.py

Phase B - Reinforcement Header Extraction

Detect section titles on SEC TEXT layer. 24 raw occurrences; 18 unique marks. Duplicate marks (B3, B12, B13, B18) = multiple detail sketches, not duplicate beams.

Modules: src/reinforcement/header_extractor.py

Phase C - Header Grid Segmentation

Cluster headers into 4 rows by Y; assign column ownership cells with midpoint boundaries. Outputs: beam_cells.json, beam_cells_validation.json.

Modules: src/grid/beam_cell_builder.py

Sketch Debug Detection

Detect reinforcement sketches per header occurrence (1 beam = N sketches). 38 sketches across 18 beams. Outputs: beam_sketches_debug.json, debug DXF.

Modules: src/grid/beam_sketch_debug_detector.py, src/geometry/geometry_graph.py

Phase C.5 - Sketch Ownership

Assign each sketch to nearest same-mark header occurrence by centroid distance. Outputs: header_occurrences.json, sketch_ownership.json, validation, debug DXF.

Modules: src/grid/sketch_ownership_builder.py, header_occurrence_exporter.py

Phase C.5.1 - Audit Metrics

Add distance_mm and confidence (HIGH/MEDIUM/LOW) per owned sketch. No assignment changes.

Modules: src/grid/sketch_ownership_auditor.py

Phase C.5.2 - Audit Refinement

Separate ownership errors from layout warnings. Long distance (>10m) = WARNING only. Ownership PASS; audit PASS_WITH_WARNINGS (B17_S2 at 12365 mm).

Modules: src/grid/sketch_ownership_validator.py

4. Key Design Concepts

- 1 beam = N sketches: multiple header labels for one mark are intentional.
- Occurrence-level ownership: B3_H1 and B3_H2 own different sketches.
- Large header-to-sketch distance is normal (sections drawn below titles).
- Debug DXF overlays on dedicated layers for AutoCAD visual verification.
- Each phase produces validated JSON before the next phase runs.

5. CLI Commands

Run from Version2/ with: \$env:PYTHONPATH="."

Script	Phases	Key outputs
docs/run_phases_abc.py	A, B, C	framing, headers, cells
docs/run_beam_cells_debug.py	C debug	beam_cells_debug.*
run_beam_sketches_debug.py	Sketches	beam_sketches_debug.*
run_sketch_ownership.py	C.5-C.5.2	ownership, validation, DXF

6. Output Files (data/output/)

JSON artifacts

- framing_beams.json, framing_validation.json
- reinforcement_headers.json, reinforcement_header_validation.json
- beam_cells.json, beam_cells_validation.json
- beam_sketches_debug.json, beam_sketches_debug_validation.json
- header_occurrences.json, sketch_ownership.json
- sketch_ownership_validation.json

DXF debug overlays

- beam_cells_debug.dxf - DEBUG_BEAM_CELLS
- beam_sketches_debug.dxf - sketch bboxes
- sketch_ownership_debug.dxf - DEBUG_SKETCH_OWNERSHIP (leaders, distances, WARNING)

7. Distance Distribution (C.5.2 audit)

Distance band	Count
0-1500 mm	0
1500-4000 mm	16
4000-8000 mm	11
8000+ mm	11

8. Not Yet Implemented (Phase D+)

- Phase D - annotation ownership within sketches
- Reinforcement steel quantity parsing
- Cross-linking framing spans to reinforcement details
- Estimator / billing calculations

Next step: Phase D after review and approval of ownership validation.

